

Natural Selection

Darwin's mechanism of evolution by natural selection consists of four observations about the natural world: overproduction, variation, limits on population growth and differential reproductive success.

1. Overproduction

- Each species produces more offspring than will survive to maturity.
- Natural populations have the reproductive potential to continuously increase their numbers over time.
- For example, if each breeding pair of elephants produces 6 offspring during 90-year life-span, in 750 years a single pair of elephants will have given rise to a population of 19 million! Yet elephants have not overrun the planet.

2. Variation

- The individuals in population exhibit variation.
- Each individual has unique combination of traits such as size, color, and ability to tolerate harsh environments.
- Some traits improve the chances, whereas others do not.
- It is important to remember that the variation necessary for evolution by natural selection must be inherited so that it can be passed to offspring.

3. Limits on population growth

- There is only so much food, water, light, growing space and so on available to population and organisms compete with each other for limited resources available to them.
- Because there are more individuals than the environment can support, not all of the offspring will survive to reproductive age.

- other limits on population growth include predators and diseases.

4 Differential reproductive success

- Those individuals that possess the most favorable combination of characteristics (those that make individuals better adapted to their environment) are more likely to survive, reproduce and pass their traits to the next generation.
- Because offspring tend to resemble their parents, the next generation obtains the parent's heritable traits.
- Sexual reproduction is the key to natural selection.
- The best-adapted individuals are those that reproduce most successfully, whereas less-fit individuals die prematurely or produce fewer or inferior offspring.
- Over time, enough changes may accumulate in geographically separated populations (often with slightly different environments) to produce new species.

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